

CS6910: Fundamentals of Deep Learning
Programming Assignment 1

Comparative Analysis of Neural Network Training Techniques

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1 Task 1: Optimization Methods Comparison

Objective: Compare five weight update rules for neural network training in pattern mode.

Methods Tested: SGD, Momentum, AdaGrad, RMSProp, and AdaM.

Convergence Behavior (from Training Error Curve)

- **SGD:** Shows a very fast initial decrease in training error, but plateaus early (around ~ 77 epochs). This indicates a limited ability to escape shallow minima.
- **Momentum:** Improves over plain SGD, but still struggles with stable convergence. The training curve exhibits oscillatory behavior before stabilizing.
- **RMSProp:** Displays unstable behavior with large fluctuations in training error. This suggests high sensitivity to the learning rate or noisy gradient updates.
- **Adam:** Demonstrates smooth and stable convergence, but does not achieve the best final test accuracy.
- **AdaGrad:** Converges slowly but steadily, and achieves the highest test accuracy (57.8%). This indicates better generalization despite slower learning.

Task 1: Confusion Matrix Analysis

Observations

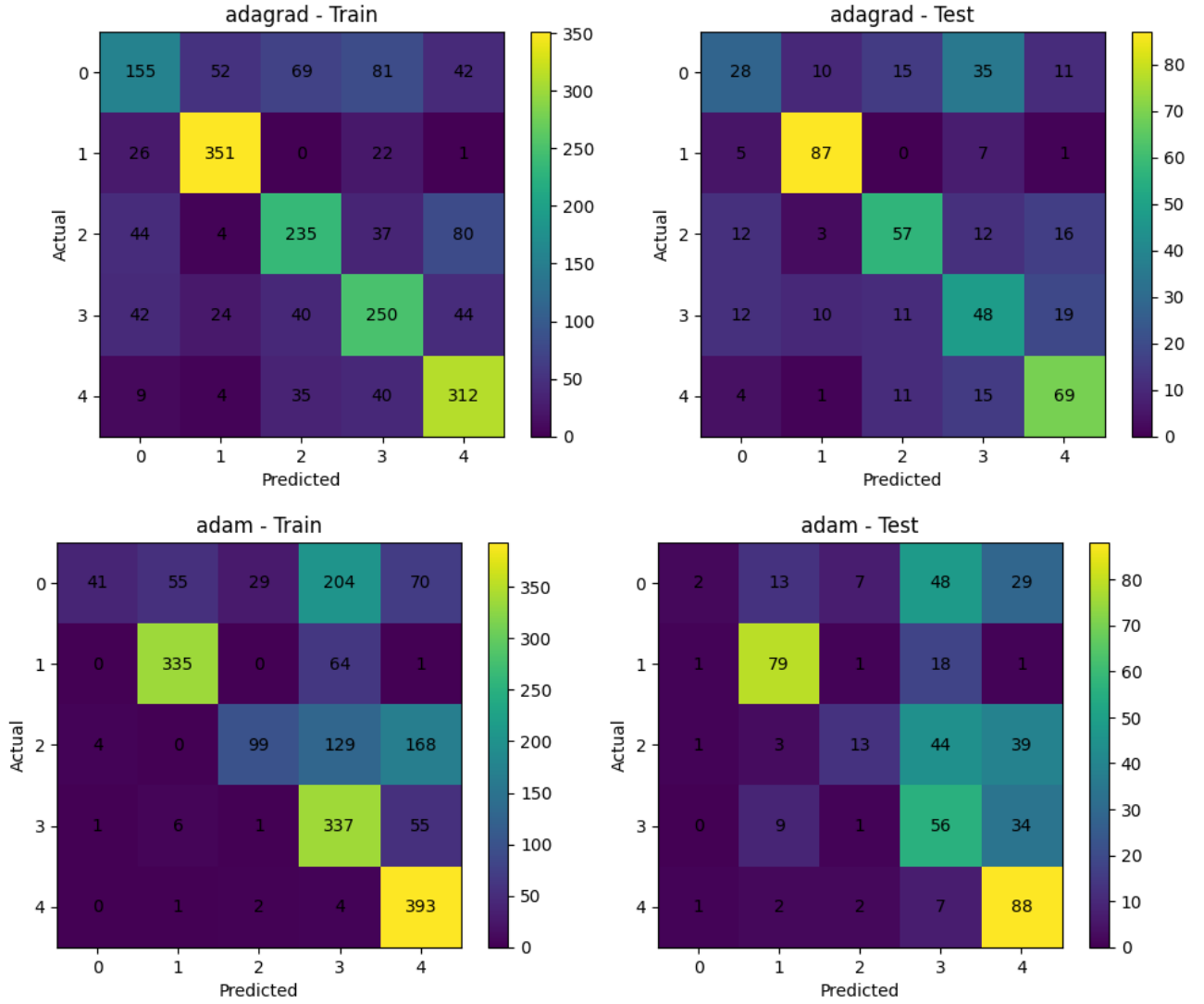
- **AdaGrad** generalizes the best, even though it required 200 epochs to converge. This suggests that an adaptive learning rate per parameter helps prevent overfitting.
- **Adam** exhibits good training stability but weaker generalization compared to AdaGrad.
- **RMSProp** and **Momentum** perform poorly on test data, indicating either overfitting or unstable parameter updates.
- The gap between training stability and test accuracy highlights that faster convergence does not necessarily guarantee better generalization.

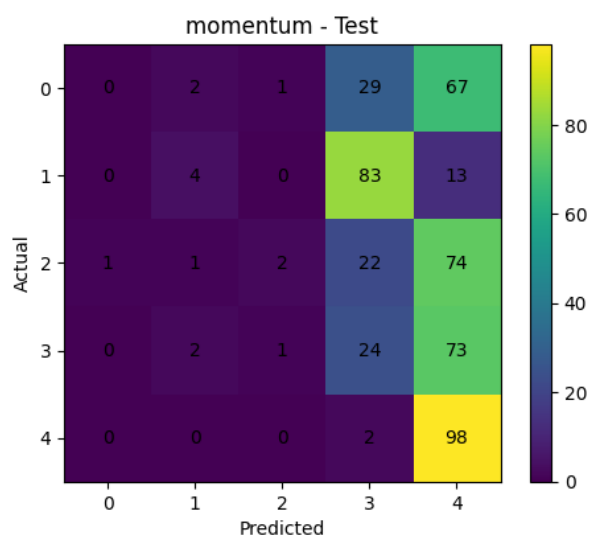
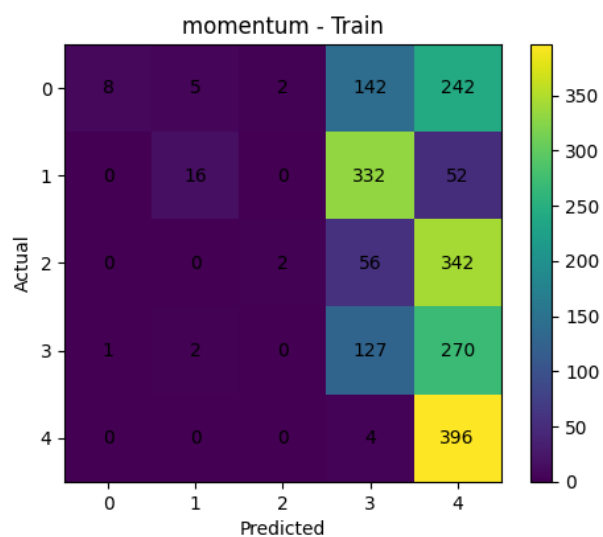
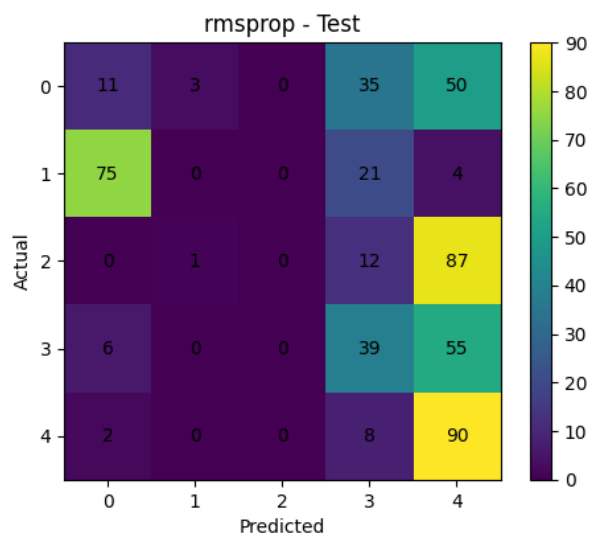
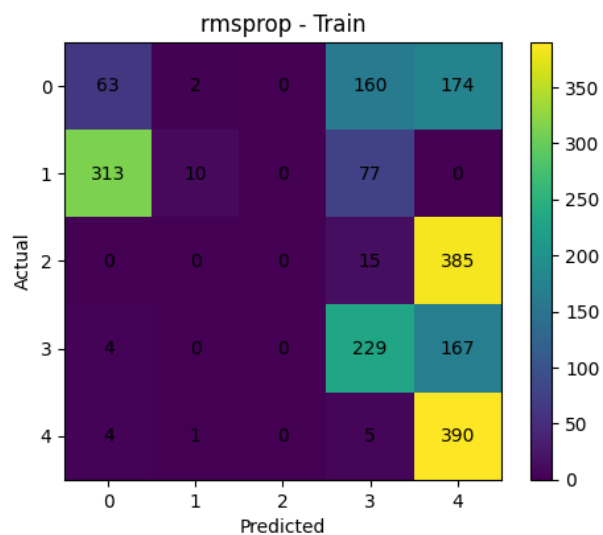
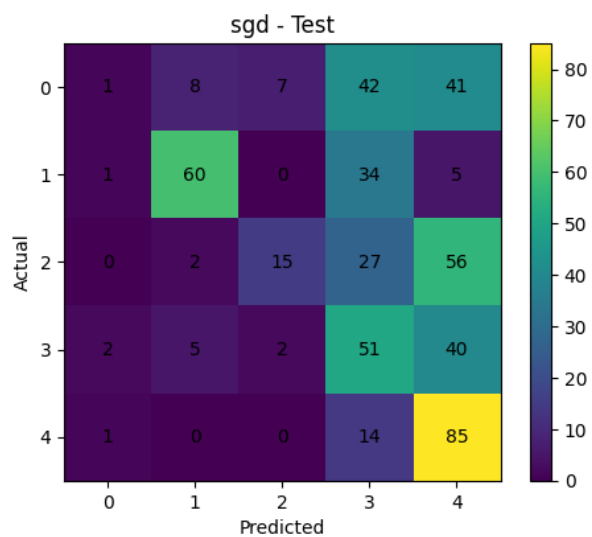
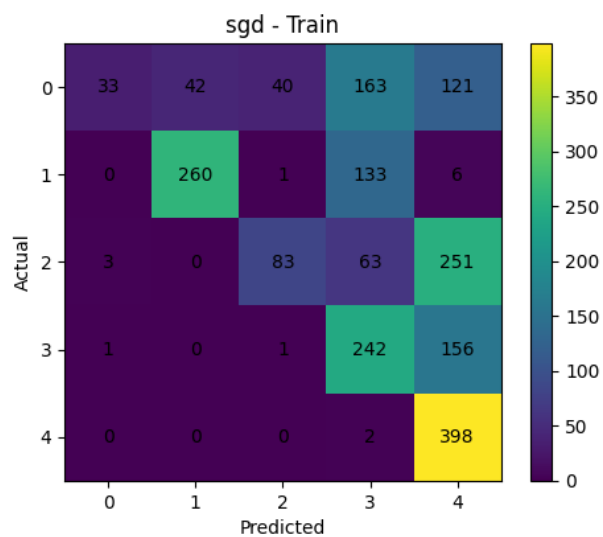
Insights from Confusion Matrices

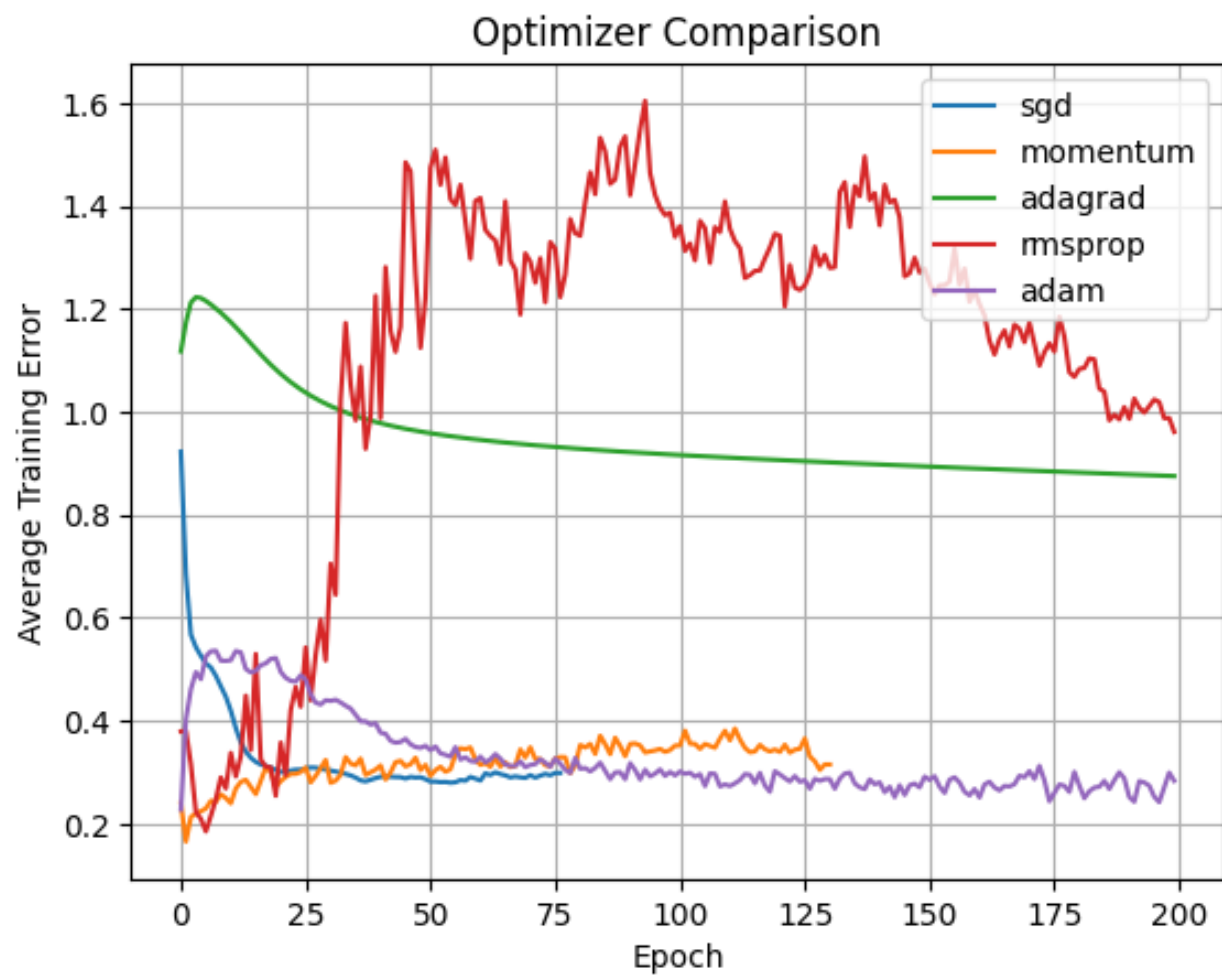
- Certain classes show consistent misclassification across all optimizers.
- **AdaGrad** exhibits relatively stronger diagonal dominance in the test confusion matrix, indicating more accurate class-wise predictions.

Table 1: Task 1 Optimization Results

Optimizer	Epochs	Test Accuracy (%)
AdaGrad	200	57.92%
Adam	200	47.70%
SGD	77	42.48%
RMSProp	200	28.06%
Momentum	130	25.65%







2 Task 2: Normalization Methods Comparison

Observations

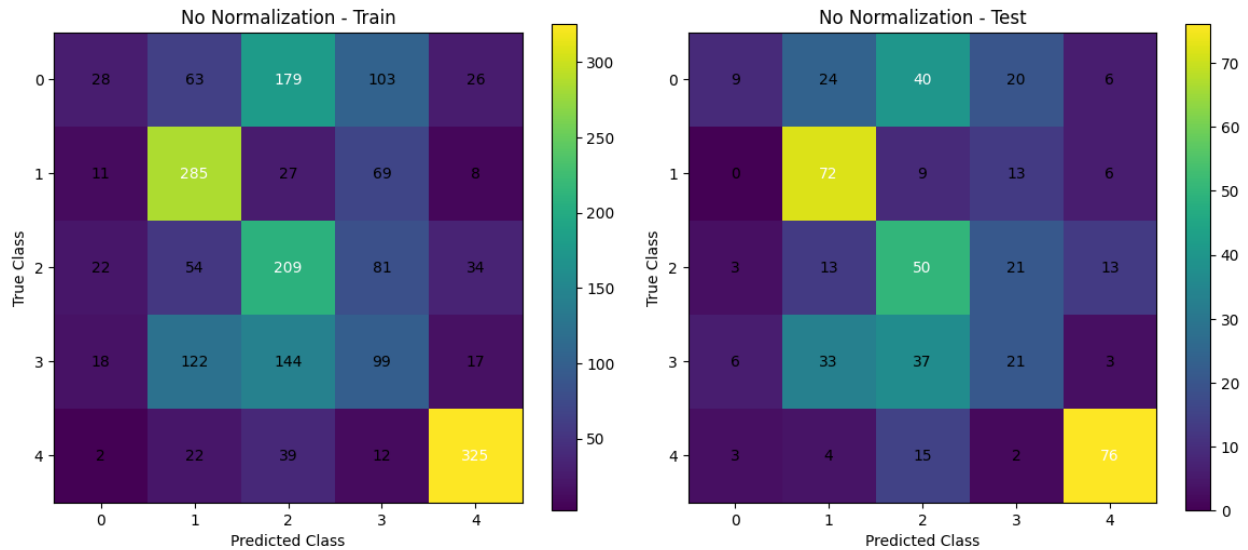
- **Layer Normalization** converges the fastest (around 32 epochs). This indicates that it stabilizes internal activations quickly during training.
- **Batch Normalization** achieves the highest test accuracy (52.6%), even though:
 - it takes longer to converge compared to Layer Normalization, and
 - it exhibits mild oscillations in the training curve.
- **No Normalization** performs the worst in terms of generalization. This suggests that internal covariate shift adversely affects training dynamics.

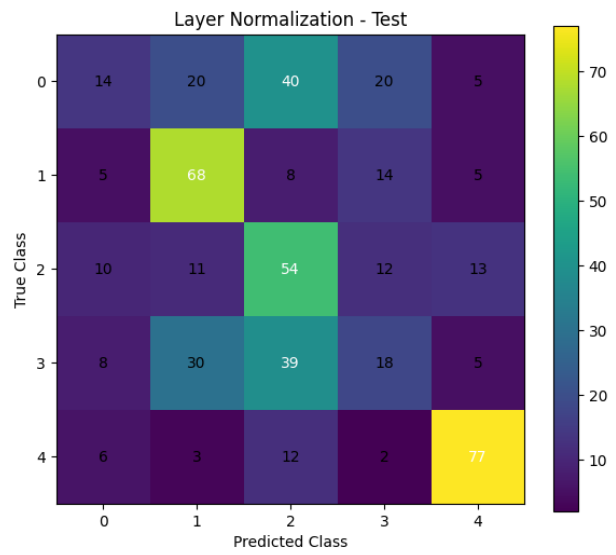
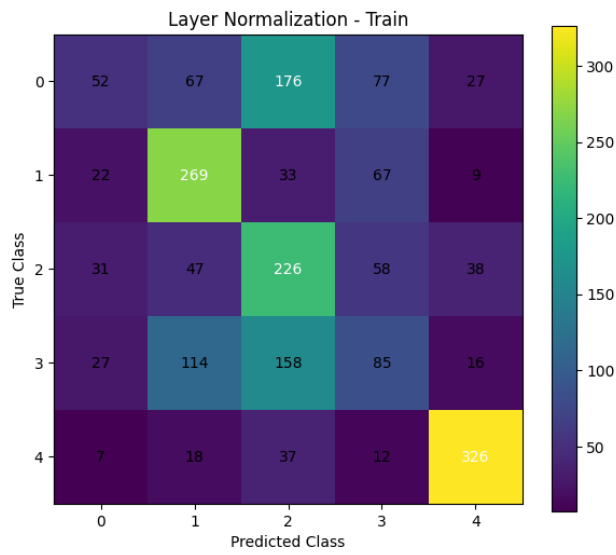
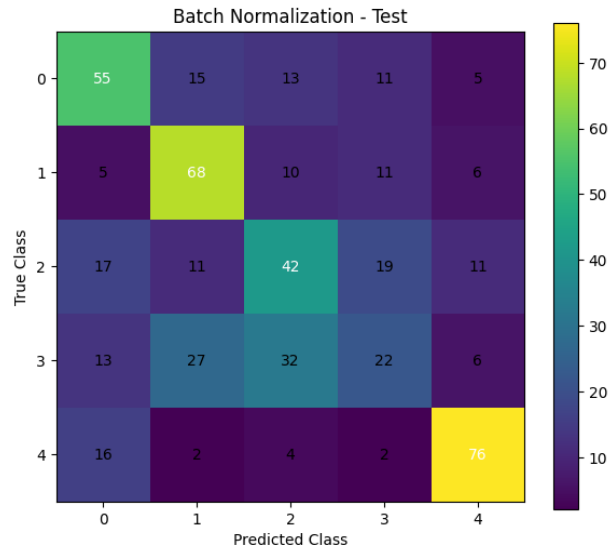
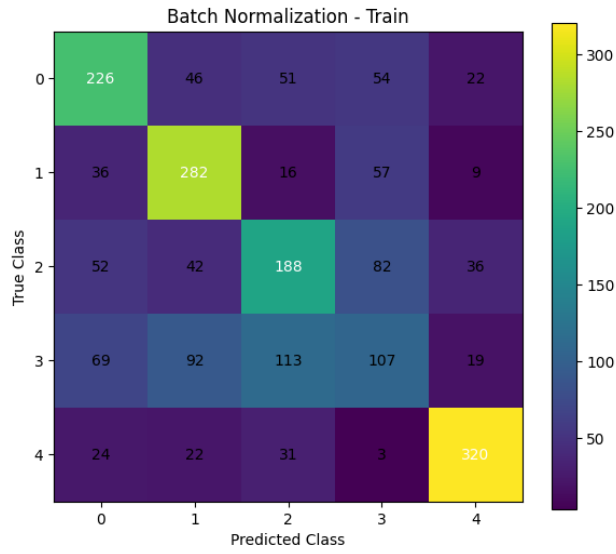
Training Curve Analysis

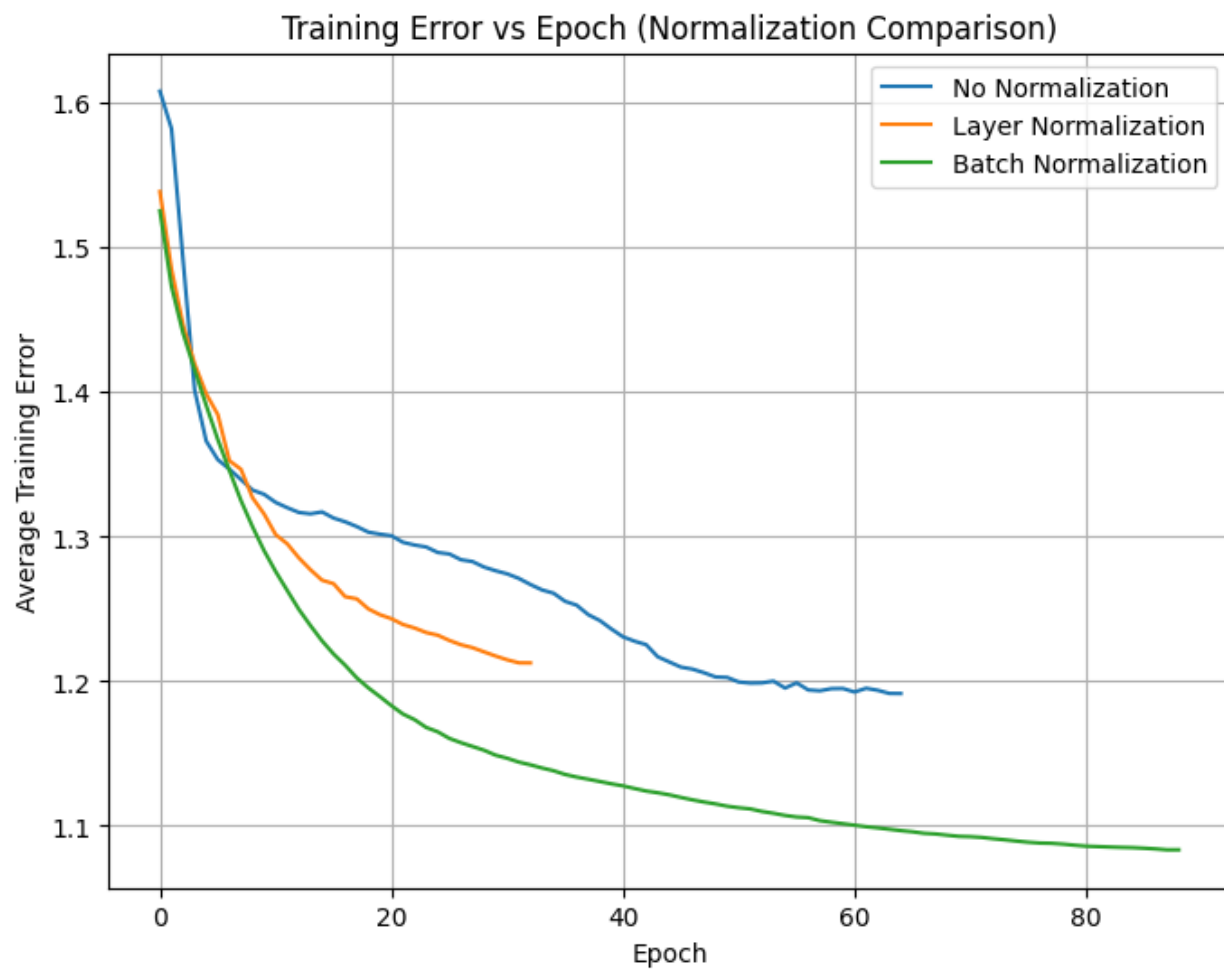
- Batch Normalization reduces training error more effectively during later epochs.
- Layer Normalization plateaus early, limiting further improvement.

Table 2: Task 2 Normalization Results

Normalization Method	Epochs to Converge	Test Accuracy (%)
Batch Normalization	88	52.6%
Layer Normalization	32	46.2%
No Normalization	64	45.6%







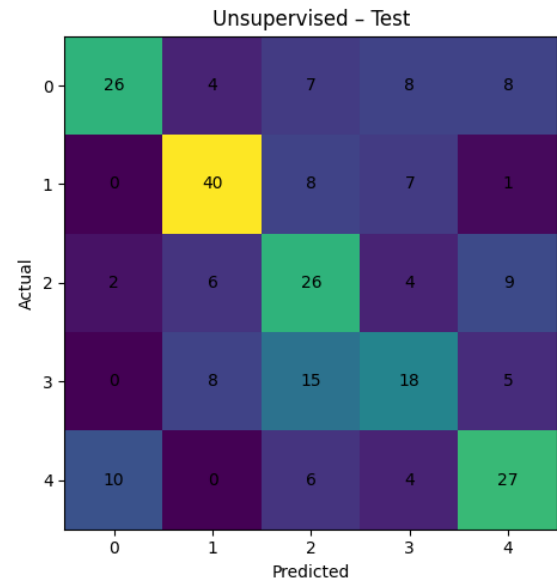
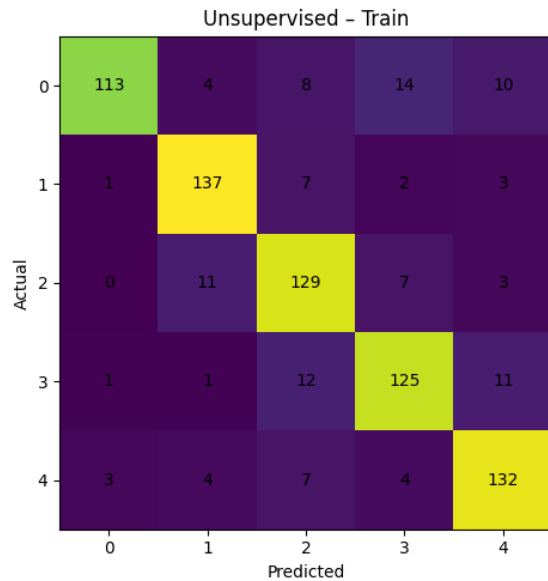
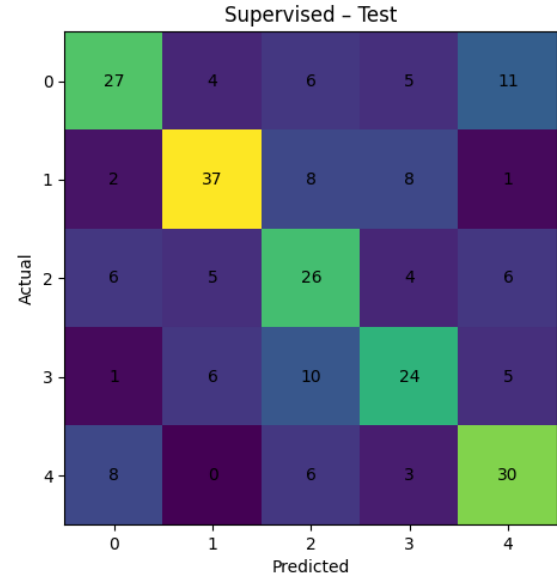
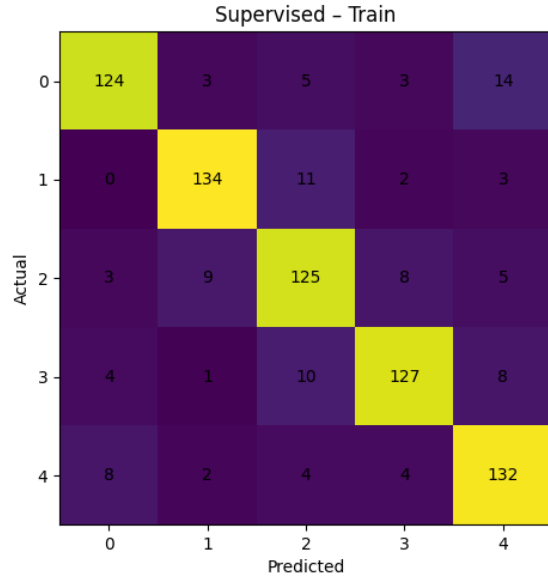
3 Task 3: Semi-Supervised Learning

Objective: Evaluate the benefit of unsupervised pre-training using a 3-layer stacked autoencoder when labeled data is limited.

- **Supervised Only Accuracy:** 57.83%
- **Pre-trained + Fine-tuned Accuracy:** 55.02%

Observations

1. Both models achieve comparable performance on the test set, with **Supervised Only** = 57.83% and **Pre-trained + Fine-tuned** = 55.02%, indicating stable and reasonable generalization.
2. The **Supervised Only** model slightly outperforms the pre-trained model, suggesting that the labeled dataset is sufficiently informative for learning discriminative features.
3. The small performance gap (2%) between the two models indicates that unsupervised pretraining provides limited additional benefit in this experimental setup.
4. There is no evidence of severe overfitting, as both models maintain balanced performance on unseen data.
5. The reconstruction-based pretraining objective may not be well aligned with the classification objective, which explains why the pre-trained model does not significantly outperform the supervised model.
6. Overall, in this experiment, **direct supervised learning is sufficiently effective**, and stacked autoencoder pretraining does not result in a substantial improvement in test accuracy.



4 Conclusion

This assignment evaluated the effects of optimization methods, normalization techniques, and semi-supervised learning on neural network training. Among optimizers, **AdaGrad** achieved the best generalization despite slower convergence, showing that stable parameter updates matter more than speed. In normalization experiments, **Batch Normalization** produced the *highest test accuracy*, while **Layer Normalization** had the *fastest convergence*. Semi-supervised pretraining using stacked autoencoders provided limited benefit, as direct supervised learning performed comparably. Overall, the results highlight that faster convergence does not necessarily lead to better generalization, and training strategies must be chosen based on task alignment rather than convergence speed alone.